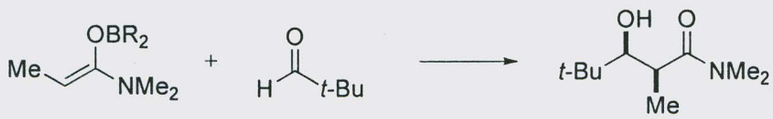
Organic Cume
September 10, 2010

Define and give an example of each of the following terms:

1. *A* value
2. $A^{1,3}$-strain
3. The anomeric effect
4. Atropisomer
5. Chirality axis
6. Diastereotopic faces
7. Racemate
8. Selectivity factor, **s**
9. Synclinal conformation
10. Stereogenic center

Answer the following:

11. Draw digraph and use the CIP system to show that a phenyl substituent has a higher CIP priority than a cyclohexyl substituent.

12. In the following hypothetical reaction between an enolate and an aldehyde, determine which face of each component reacted to form the product, and state whether the relative topicity is *lk* or *ul.*